

Boyce-Codd Normal Form (BCNF) Relational Database Design

> **Definition:** A relation is in **Boyce-Codd Normal Form (BCNF)** (also known as 3.5NF) if and only if for **every non-trivial functional dependency $X \rightarrow Y$** , **X is a strict Super Key**.

1. Detailed Technical Explanation

BCNF is a stricter version of 3NF that handles cases where a relation has **multiple overlapping candidate keys**.

3NF vs BCNF Comparison:

| Property | 3NF | BCNF |

| :--- | :---

Conversion Example to BCNF

Relation ADVISOR(Student_ID, Subject, Advisor_Name)

- **A**

Relation Status:

- Is it

BCNF Decomposition:

1. **AD**

2. Core Concepts & Memory Keywords

- **Strict Super Key Rule:** Every determinant in a functional dependency must be a super key.

- **C**

3. Must-Write Points for Exams

- BCNF eliminates all redundancy resulting from functional dependencies.

- Eve

4. Quick Recall Flow

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3NF R